

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 7 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTCVE705A - ENGINEERING ECONOMICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 The Study of Engineering Economics mainly deals with a) Production onlyb) Demand and supply, cost, value of moneyc) Machine designd) Thermodynamics
- 2 When demand is perfectly elastic, the elasticity value is:a) Zerob) Onec) Infinityd) Negative
- 3 What is the most important concept in engineering economya) Demand and supplyb) Types of costsc) Time value of moneyd) Balance sheets
- 4 Continuous compounding interest formula is:a) $A = P(1+i)^n$ b) $A = Pe^{rt}$ c) $A = P/\eta d) A = P - r$
- 5 Parametric estimation is based on:a) Random trialsb) Statistical correlation of parameterc) Market demandd) Annual budgeting
- 6 Depreciation is caused due to:a) Wear & tearb) Obsolescencetc) Accidentsd) All of the above
- 7 A systematic allocation of the cost of a tangible assest over its useful life is calleda) Taxationb) Depreciationc) Investment evaluationd) Working capital
- 8 Which method of depreciation gives maximum depreciation in the first year?a) Straight lineb) Double decliningc) Sinking fundd) Unit of production
- 9 What does the gradient factor $(A/G, i, n)$ convert?a) A gradient series into present worthb) A gradient series into future worthc) A gradient series into uniform annual seriesd) A uniform series into gradient series
- 10 The financial ratio useful for liquidity is:a) Debt ratio b) Current ratio c) Return ratio d) Activity ratio
- 11 Capital recovery factor (A/P) is used to find:a) Annual equivalent of present valueb) Future value of annuityc) Present value of gradientd) Depreciation
- 12 The depreciation balance method is also known asa) Straight line methodb) Written down value methodc) Sum of the year's digitald) Unit of production method

Q2. Solve the following.

- A) Explain demand and supply curves with neat diagram.

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- B) Discuss the classification of cost in detail and develop a cost - time relation ship with an example. 6
- Q3. Solve the following.
- A) Drive the formula for uniform series present worth(P/A).also give engineering applications. 6
- B) Differentiate between nominal interest rate and effective interest rate. derive the formula for effective interest rate. 6
- Q4. Solve Any Two of the following.
- A) Explain management accounting. how does it support decision making in engineering projects. 6
- B) Explain lease financing with advantages, limitation and evaluation criteria. 6
- C) What are financial accounting principles? explain any four basic accounting principles with examples. 6
- Q5. Solve Any Two of the following.
- A) Explain the concept of cost of capital. how it is used in making financing decision for projects? 6
- B) What is taxation in engineering economics? explain different types of taxes applicable to engineering projects 6
- C) Explain in detail the concept of risks in project financing and how they are managed. 6
- Q6. Solve Any Two of the following.
- A) Explain any two depreciation methods in detail. compare their effect on book value and tax liability. 6
- B) What is profit and loss statement. explain its structure and importance in evaluating profitability. 6
- C) What is international finance? explain factors influencing international financial flow. 6

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